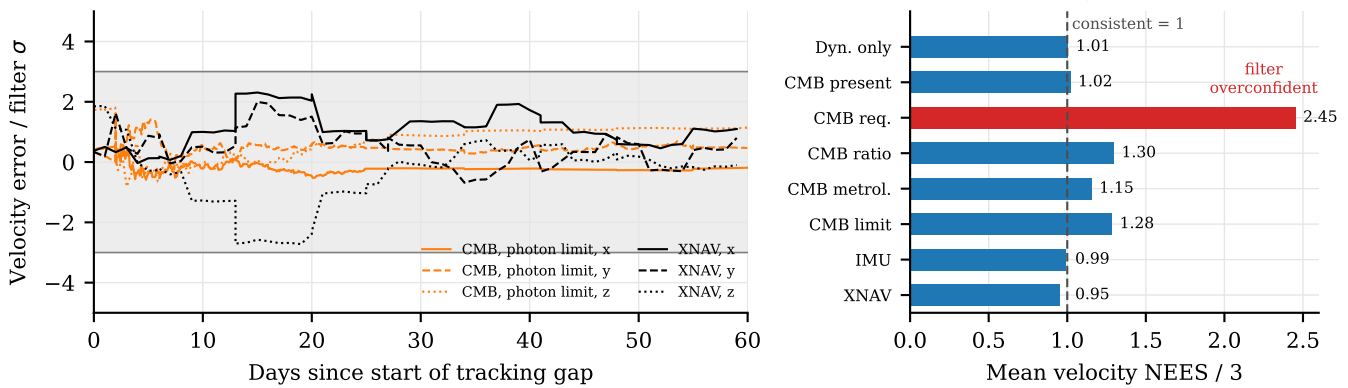
(a) Normalized error per axis (one seed)
Velocity error / filter σ
4
2
0
−2
−4
0
10
20
30
40
50
60
Days since start of tracking gap
CMB, photon limit, x
CMB, photon limit, y
CMB, photon limit, z
XNAV, x
XNAV, y
XNAV, z
(b) Consistency per case (6 seeds)
consistent = 1
Dyn. only
CMB present
CMB req.
CMB ratio
CMB metrol.
CMB limit
IMU
XNAV
1.01
1.02
2.45
1.30
1.15
1.28
0.99
0.95
filter
overconfident
0.0
0.5
1.0
1.5
2.0
2.5
Mean velocity NEES / 3